

lab9-ao

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```
#import data and input into name
candy_file<-"candy-data.csv"
candy=read.csv(candy_file,row.names = 1)

head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

Answer Q1 There are 85 total candy types in the dataset.

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset? **Answer Q2** There are 38 fruity candy types.

```
sum(candy$fruity)
```

```
[1] 38
```

```
#example given to calculate amount of  
#perferred twix candy  
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
#alternative code from previous code (twix)  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>  
  filter(row.names(candy)==  
         "Twix") |>  
  select(winpercent)
```

```
      winpercent  
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value? **Answer Q3** : Peanut butter M&M's are my favorite with a winpercent of 71.47

```
candy["Peanut butter M&M's", ]$winpercent
```

```
[1] 71.46505
```

Q4. What is the winpercent value for “Kit Kat”? **Answer Q4** : The winpercent is 76.77 for Kit Kat’s.

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”? **Answer Q5** : Tootsie Roll Snack Bars have a winpercent of 49.65

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

```
#install function of skimr and
#we are given mean,sd, rates
library("skimr")

skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

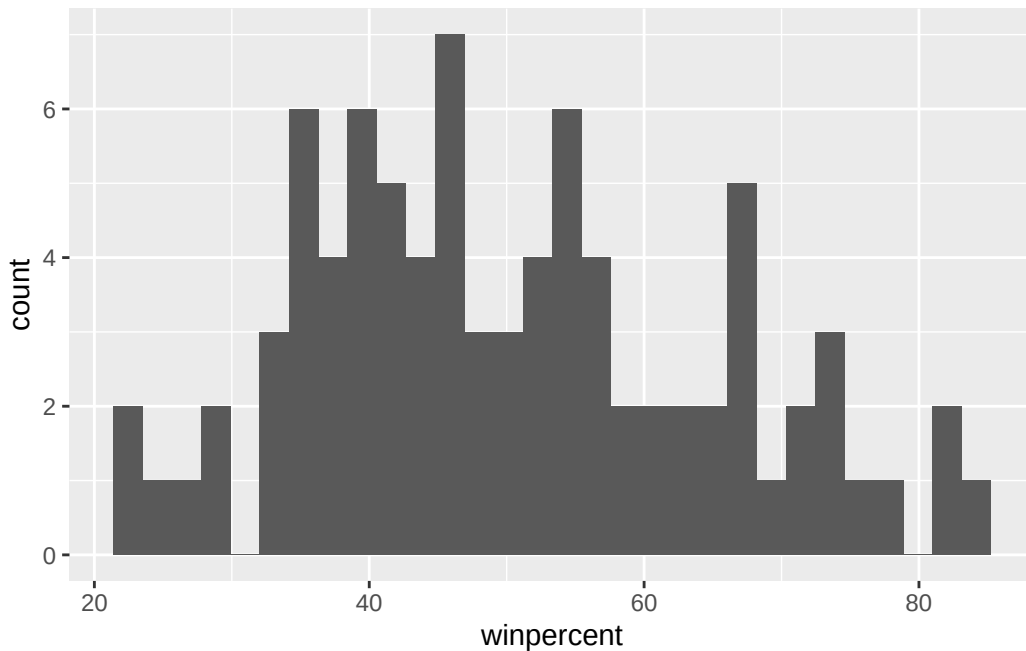
Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset? **Answer Q6** Yes, all variables are between 0-1 in data except winpercent is ranging from 0-100 based on skimr function output.

Q7. What do you think a zero and one represent for the candy\$chocolate column? **Answer Q7** Zero shows candy without any chocolate and 1 signifies that the candy does have chocolate.

Q8. Plot a histogram of winpercent values using both base R and ggplot2. **Answer Q8:** The code below would use geom_hist() and assign x axis to winpercent.

```
library(ggplot2)
ggplot(candy,aes(x=winpercent))+geom_histogram()
```

``stat_bin()` using `bins = 30`. Pick better value `binwidth`.`



Q9. Is the distribution of winpercent values symmetrical? **Answer Q9:** No, the distribution shows to be right-skewed.

Q10. Is the center of the distribution above or below 50%? **Answer Q10 :** The center is below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy? **Answer Q11:** The average is ranked higher for chocolate candy (61) than fruit candy (44).

```
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

Q12. Is this difference statistically significant? **Answer Q12 :** Yes, based on the t test demonstrating the p value (2.871e-08) is much lesser than zero.

```
t.test(candy$winpercent[as.logical(candy$chocolate)],
       candy$winpercent[as.logical(candy$fruity)])
```

Welch Two Sample t-test

```
data: candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fruity)]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Q13. What are the five least liked candy types in this set? **Answer Q13** : They are the following : Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, Jawbusters.

```
#least liked
head(candy[order(candy$winpercent),],n=5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0

	crispedricewafer	hard bar	pluribus	sugarpercent	pricepercent	
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

```
#most liked
tail(candy[order(candy$winpercent),],n=5)
```

```
chocolate fruity caramel peanutyalmondy nougat
```

Snickers	1	0	1	1	1
Kit Kat	1	0	0	0	0
Twix	1	0	1	0	0
Reese's Miniatures	1	0	0	1	0
Reese's Peanut Butter cup	1	0	0	1	0

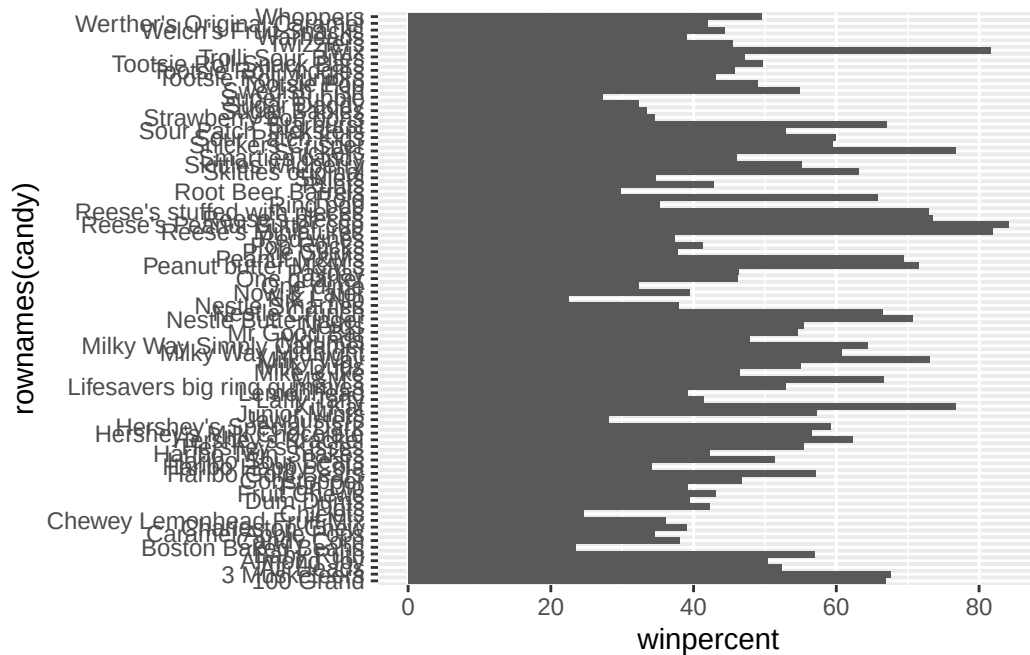
	crisped	rice	wafers	hard	bar	pluribus	sugar	percent
Snickers		0	0	1		0		0.546
Kit Kat		1	0	1		0		0.313
Twix		1	0	1		0		0.546
Reese's Miniatures		0	0	0		0		0.034
Reese's Peanut Butter cup		0	0	0		0		0.720

	price	percent	win	percent
Snickers	0.651	76.67378		
Kit Kat	0.511	76.76860		
Twix	0.906	81.64291		
Reese's Miniatures	0.279	81.86626		
Reese's Peanut Butter cup	0.651	84.18029		

Q14. What are the top 5 all time favorite candy types out of this set? **Answer Q14:** Based on the previous code output, they are the following : Snickers, Kit Kat, Twix, Reese's Miniatures, Reese's Peanut Butter cup.

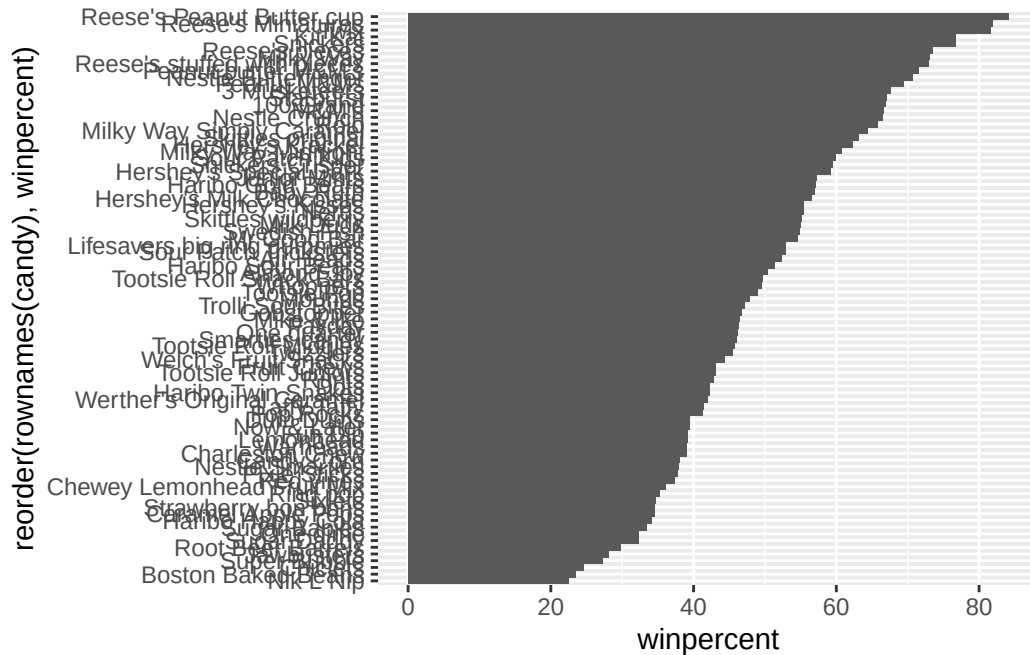
Q15. Make a first barplot of candy ranking based on winpercent values. **Answer Q15 :** The code will include the x axis to be assigned with winpercent data, y axis are the candies.

```
#bar plot before ranking based on winpercent data
ggplot(candy,aes(x=winpercent,y=rownames(candy)))+
  geom_col()
```



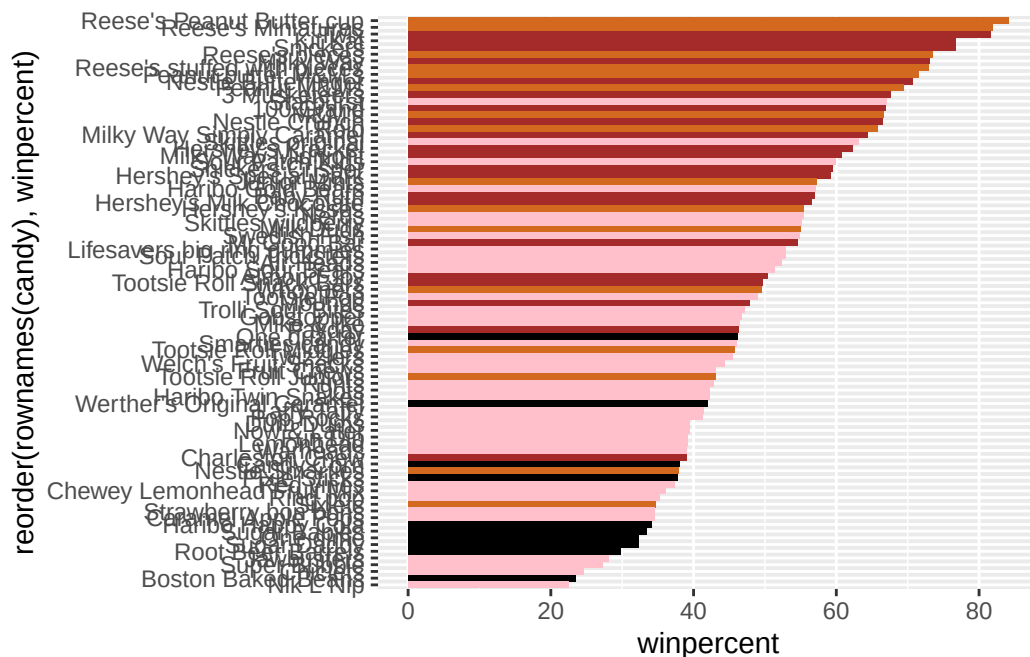
Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?
Answer Q16 : `rownames()` was placed inside the function `reorder()` then `winpercent` placed after to be sorted by winpercent.

```
#sort bar plot by winpercent using reorder()
ggplot(candy,aes(x=winpercent,y=
                 reorder(rownames(candy),
                         winpercent)))+
geom_col()
```

```
#set up colors per vector for future plots
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"

#generate barplot with the colors assigned
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```



Q17. What is the worst ranked chocolate candy? **Answer Q17** : Nik L Nip

```
candy[as.logical(candy$winpercent),] |>
  arrange(winpercent) |>
  head(1)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer	hard
Nik L Nip	0	1	0	0	0	0	0
	bar	pluribus	sugarpercent	pricepercent	winpercent		
Nik L Nip	0	1	0.197	0.976	22.44534		

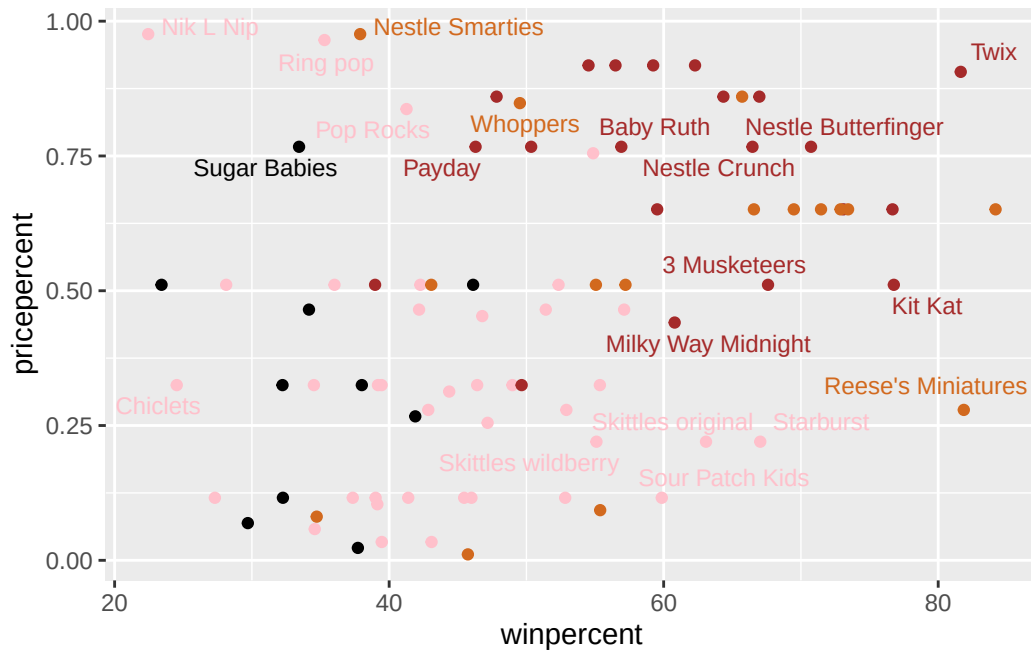
Q18. What is the best ranked fruity candy? **Answer Q18** : Starburst

```
candy[as.logical(candy$fruity), ] |>
  arrange(desc(winpercent)) |>
  head(1)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer	hard
Starburst	0	1	0	0	0	0	0
	bar	pluribus	sugarpercent	pricepercent	winpercent		
Starburst	0	1	0.151	0.22	67.03763		

```
library(ggrepel)

#plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck? **Answer Q19** : Tootsie Roll Midgies

```
ord<-order(candy$winpercent/candy$pricepercent,decreasing = TRUE)
head(candy[ord,],1)
```

	chocolate	fruity	caramel	peanut	almond	nougat	
Tootsie Roll Midgies	1	0	0	0	0	0	
	crisp	edrice	wafer	hard	bar	pluribus	sugarpercent
Tootsie Roll Midgies	0	0	0	1	0.174		
	pricepercent	winpercent					
Tootsie Roll Midgies	0.011	45.73675					

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular? **Answer Q20** : Nik L Nip, Nestle Smarties, Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate are the 5 most expensive candies and least popular. Nik L Nip is the least popular.

```
#most expensive
ord<-order(candy$pricepercent,decreasing = TRUE)
top5candies<-head(candy[ord,],5)
head(top5candies)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Nestle Smarties	1	0	0		0	0
Ring pop	0	1	0		0	0
Hershey's Krackel	1	0	0		0	0
Hershey's Milk Chocolate	1	0	0		0	0

	crisped	rice	wafer	hard bar	pluribus	sugar	percent
Nik L Nip		0	0	0	1		0.197
Nestle Smarties		0	0	0	1		0.267
Ring pop		0	1	0	0		0.732
Hershey's Krackel		1	0	1	0		0.430
Hershey's Milk Chocolate		0	0	1	0		0.430

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

```
#least popular
top5candies[which.min(top5candies$winpercent)]
```

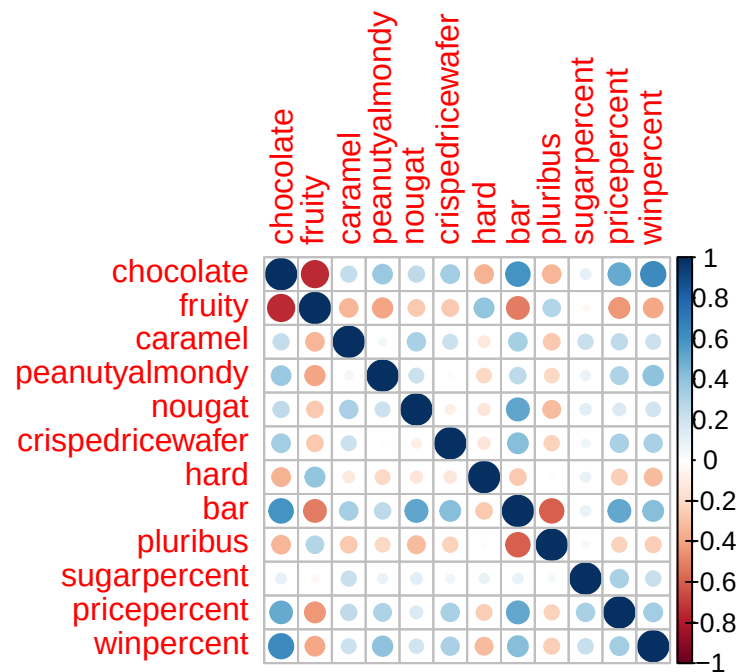
	chocolate
Nik L Nip	0
Nestle Smarties	1
Ring pop	0
Hershey's Krackel	1
Hershey's Milk Chocolate	1

```
#generate correlation structure
library(corrplot)
```

corrplot 0.95 loaded

```
cij<-cor(candy)

corrplot(cij)
```



Q21 - optional

Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Answer Q22 Chocolate and fruity are anti-correlated.

Q23. Similarly, what two variables are most positively correlated? **Answer Q23** Winpercent and bar are positively correlated

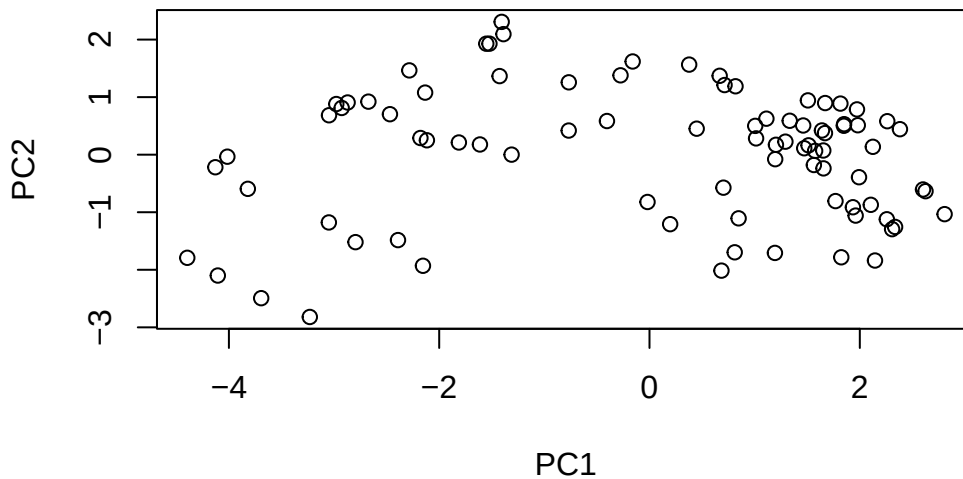
```
#input prcomp() to set scale=true argument
pca <- prcomp(candy,scale=TRUE)
summary(pca)
```

Importance of components:

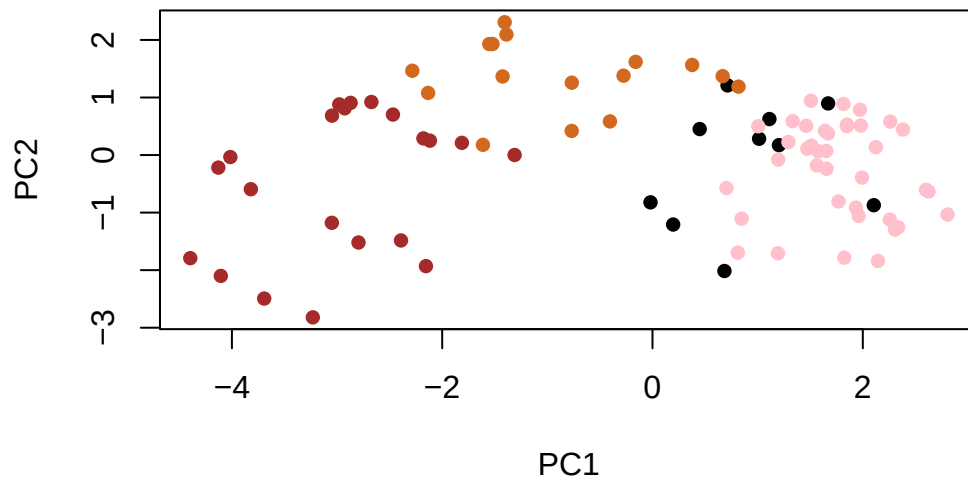
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
#plot main PCA score plot of PC1 vs PC2  
plot(pca$x[,1:2])
```



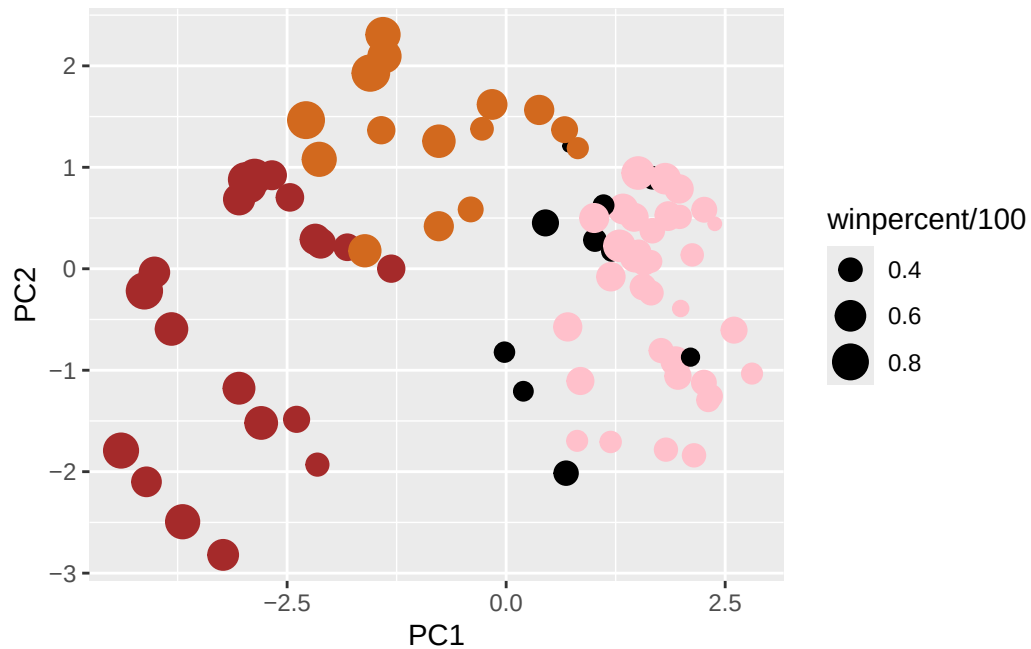
```
#add color and change plotting character  
plot(pca$x[,1:2], col=my_cols, pch=16)
```



```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```
p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=my_cols)
```

p



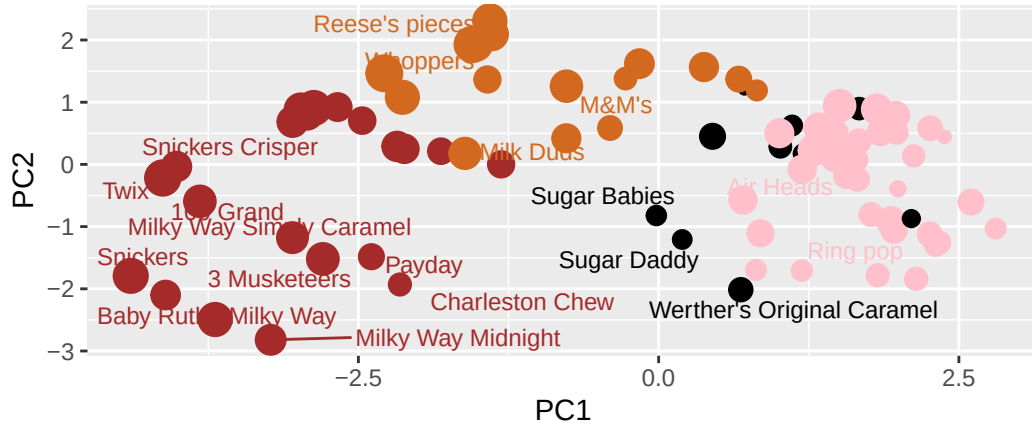
```
library(ggrepel)
#label plot with non overlapping candy names + subtitle

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type:
        chocolate bar (dark brown),
        chocolate other (light brown),
        fruity (red), other (black)",
        caption="Data from 538")
```


Halloween Candy PCA Space

Colored by type:

chocolate bar (dark brown),
chocolate other (light brown),
fruity (red), other (black)



Data from 538

```
#view more candy labels via interactive plot  
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

last_plot

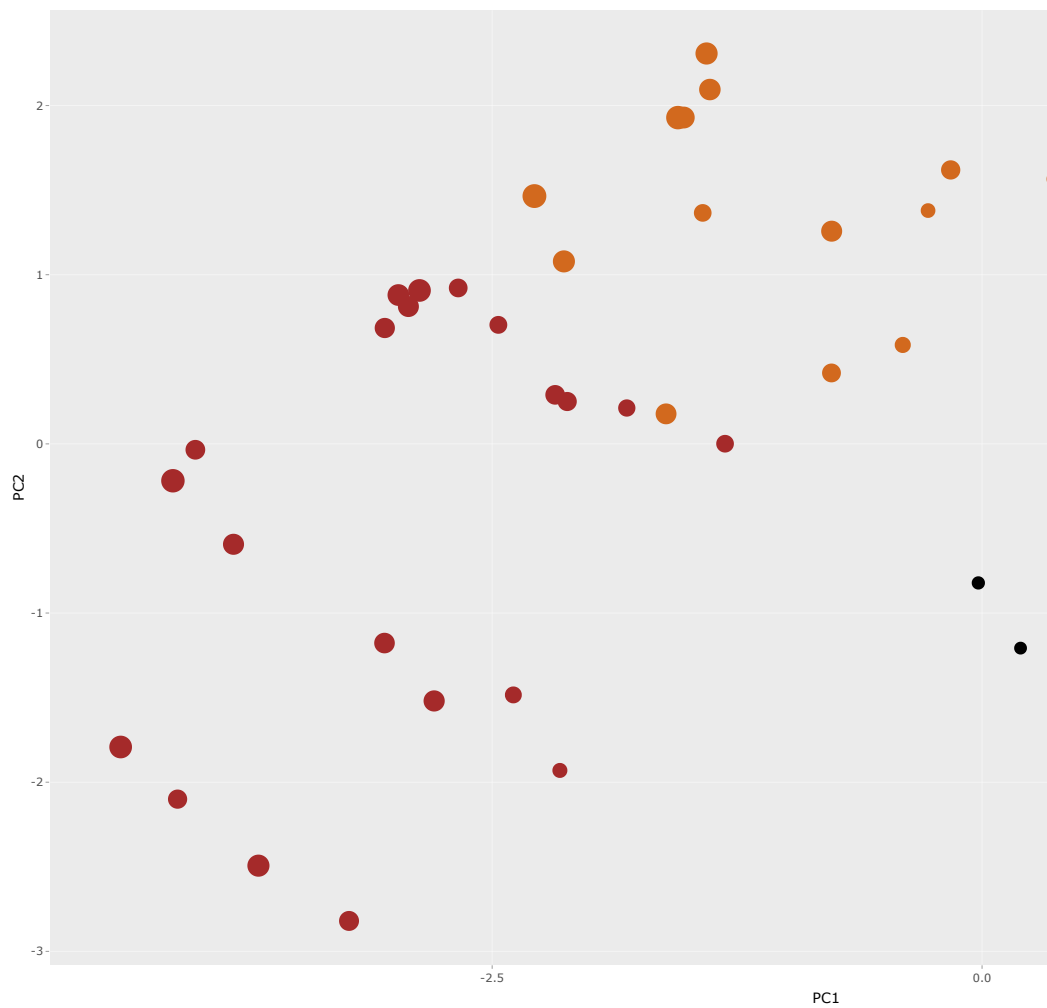
The following object is masked from 'package:stats':

filter

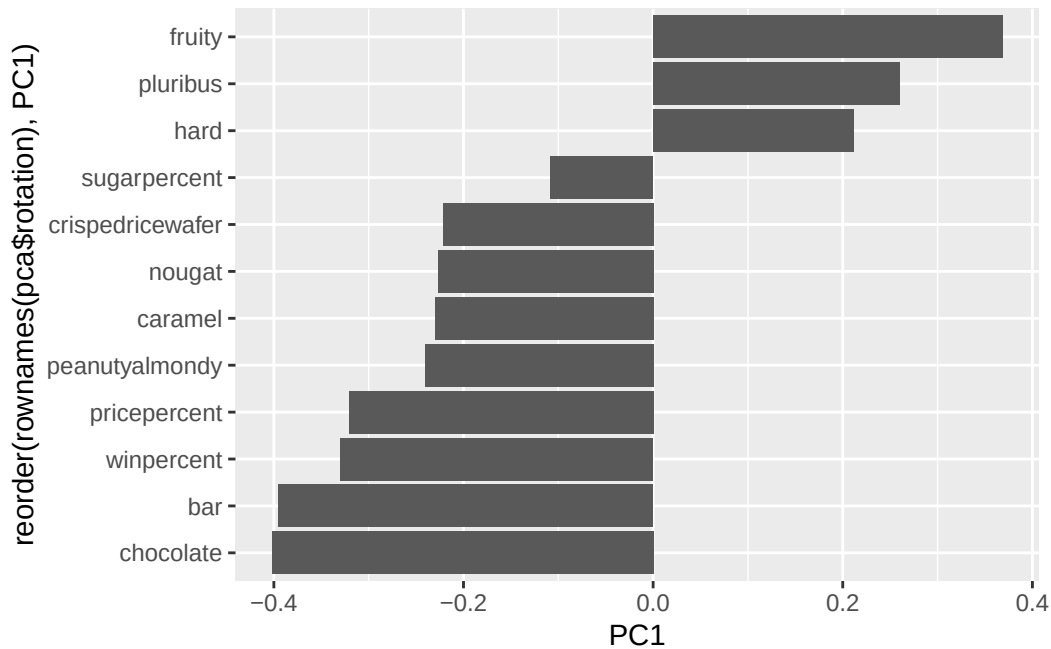
The following object is masked from 'package:graphics':

layout

```
ggplotly(p)
```



```
ggplot(pca$rotation) +
  aes(PC1, reorder(rownames(
    pca$rotation),PC1)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously? **Answer Q24** : PC1 strongly detects fruity, pluribus, and hard candy to be in the positive direction. This is due to the trend of those candies to be opposing chocolate candies which was shown in the correlation matrix (fruity and chocolate candy are anti-correlated).

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion? **Answer Q25** : A “winning candy” is categorized as a chocolate bar that is peanut/almond based based on the visualization (chocolate higher in winpercent), correlation (chocolate, bar, and peanut/almond are positively correlated) and PCA data (these variables of a “winning candy” are in negative PC1 direction).